(d) Write short notes on : 3+4

(i) Biological effect of electrical field

(ii) Faraday's cage

CSVTUonline.com

CSVTUonline.com

324635(24)

CSVTUonline.com

B. E. (Sixth Semester) Examination, Nov.-Dec., 2015

(Old Scheme)

(Elect., EEE Branch)

TESTING & COMMISSIONING of ELECTRIC EQUIPMENTS

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : *Part (a) of each question are compulsory. Attempt any two part from (b), (c) and (d). of each question.*

1. (a) Define polarization index. 2

(b) Explain the meaning of insulation resistance. How it

CSVTUonline.com

is measured for power transformer? (Insulation resistance and megger) 7

CSVTUonline.com

(c) Write and explain different type of commissioning test in a power transformer. 7

(d) What are the causes of transformer failure? 7

2. (a) What are the application of CT and PT in substation? 2

(b) Write short notes : 4+3

CSVTUonline.com

(i) On load tap changer operation

(ii) Silica gel breather

(c) What are the major difference in the design of a traction transformer in substation? 7

(d) What are the error and causes of error in CT? Explain how these error are reduced in CT. 7

3. (a) Write two reasons for sparking at brushes in a dc machine. 2

CSVTUonline.com

(b) What are various reasons for the noise? How this can be reduced in a rotating electrical machine? 7

(c) Explain the various methods of starting and drive control of induction motors. 7

CSVTUonline.com

(d) Discuss on : 3+4

(i) Degree of protection of Induction motor

(ii) Type of tests carried out on Induction motor of large capacity

(a) Why drying of motor is necessary? 2

(b) What is meant by duty on a rotating machine? How the type and capacity of motor is selected for a given application? 7

(c) Describe the methods of drying out of a rotating machine. 7

(d) Give example of standard designations and their brief description for various types of cooling. 7

(a) Define clearance in an electrical system. 2

(b) What is the Equipment earthing? How this provides safety to operator? Explain. 7

(c) Explain the principle of live line maintenance. How the safety of operator is ensured in live line working? 7

CSVTUonline.com